Linux File System Hierarchy Cheatsheet

CCEE Exam Preparation Guide

1. The Core Concept: The Inverted Tree

In Linux, everything starts at the **Root** directory, represented by a forward slash (`/`). All other directories are sub-directories of root. The structure is case-sensitive.

Essential Directories & Functions

1. The Boot & Core System

- `/` (Root): The top-level directory.

> **MCQ Exam Tip**
>
> Only the `root` user has write permissions here.

- `/boot`: Contains the Linux Kernel (`vmlinuz`), initial RAM disk image, and bootloader (GRUB) files.

> **MCQ Exam Tip**
>
> Answer `/boot` if asked where the **Kernel** resides.

- `/etc`: System-wide configuration files. *Examples:* `/etc/passwd`, `/etc/ssh/sshd_config`.

> **MCQ Exam Tip**
>
> Keywords: "Configuration," "Startup Scripts," "Settings."

2. Binaries (Executables)

- `/bin`: Essential **User** Binaries. Available to all users. *Examples:* `ls`, `cp`, `cat`, `bash`.
- `/sbin`: Essential **System** Binaries. Intended for the Administrator (Root). *Examples:* `fdisk`, `reboot`, `iptables`.

> **MCQ Exam Tip**
>
> The 's' in `sbin` stands for System/Superuser.

- `/lib`: Shared libraries required by binaries in `/bin` and `/sbin`.

3. User Data

- `/home`: Personal directories for regular users (e.g., `/home/alice`).

- `/root`: The home directory for the **Superuser (root)**.

MCQ Exam Tip

Trap Alert: Do not confuse the directory `/root` (admin's home) with `/` (the system root).

4. Variable & Temporary Data

- `/var`: Variable data that grows over time. Contains **Logs**, Mail, Spools. *Location:* `/var/log`, `/var/mail`.

MCQ Exam Tip

If the question mentions "Logs" or "Dynamic Data," the answer is `/var`.

- `/tmp`: Temporary files. Contents are usually deleted on reboot.

5. Devices & Virtual Filesystems

- `/dev`: Device files. Represents hardware as files. *Examples:* `/dev/sda` (hard drive), `/dev/tty` (terminal).
- `/proc`: Virtual filesystem residing in **RAM**. Contains runtime system info. *Examples:* `/proc/cpuinfo`, `/proc/meminfo`.

MCQ Exam Tip

It does not exist on the hard disk. It is created by the kernel at boot.

Quick Revision Table

Directory	Key Function / Keyword
`/bin`	Essential User Commands (`ls`, `cp`)
`/sbin`	Admin Commands (`reboot`, `fdisk`)
`/etc`	Configuration Files
`/var`	**Logs**, Spool, Dynamic Data
`/dev`	Device Files (Hardware)
`/proc`	Virtual Process/Kernel Info (RAM)
`/home`	User Personal Data
`/mnt`	Temporary Mount Points
`/opt`	Optional/Add-on Software

Common Exam Traps

Where are web server logs?
Wrong: `/etc/httpd` (This is the config).
Correct: `/var/log/httpd` (Logs live in `/var`).

Where is the kernel file?
Correct: `/boot`

Which directory is virtual?
Correct: `/proc` or `/sys`